15.3: Linked genes tend to be inherited together because they are located near each other on the same chromosome

The number of genes in a cell is far greater than the number of chromosomes; in fact, each chromosome (except the Y) has hundreds or thousands of genes. Genes located near each other on the same chromosome tend to be inherited together in genetic crosses; such genes are said to be genetically linked and are called **linked genes**. When geneticists follow linked genes in breeding experiments, the results deviate from those expected from Mendel's law of independent assortment.

How Linkage Affects Inheritance

To see how linkage between genes affects the inheritance of two different characters, let's examine another of Morgan's *Drosophila* experiments. In this case, the characters are body color and wing size, each with two different phenotypes. Wild-type flies have gray bodies and normal-sized wings. In addition to these flies, Morgan had managed to obtain, through breeding, doubly mutant flies with black bodies and wings much smaller than normal, called vestigial wings. The mutant alleles are recessive to the wild-type alleles, and neither gene is on a sex chromosome. In his investigation of these two genes, Morgan carried out the crosses shown in **Figure 15.9**. The first was a P generation cross to generate F1 dihybrid flies, and the second was essentially a testcross.

****Figure 15.9

Inquiry: How does linkage between two genes affect inheritance of characters?

Experiment and Results

Morgan wanted to know whether the genes for body color and wing size are genetically linked and, if so, how this affects their inheritance. The alleles for body color are b+ (gray) and b (black), and those for wing size are vg+ (normal) and vg (vestigial).

A diagram of Morgan's experiment.

Morgan mated true-breeding P (parental) generation flies, wild-type flies with black, vestigial-winged flies, to produce heterozygous F 1 dihybrids (b plus b v g plus v g), all of which are wild-type in appearance. He then mated wild-type F 1 dihybrid females with homozygous recessive males. This testcross will reveal the genotype of the eggs made by the dihybrid female. The male's sperm contributes only recessive alleles, so the phenotype of the offspring reflects the genotype of the female's eggs. Note: Although only females (with pointed abdomens) are shown, half the offspring in each class would be males (with rounded abdomens). The diagram is as follows.

P generation (homozygous). Wild type: gray body, normal wings (alleles: b plus, b plus, v g plus, v g plus) x Double black mutant: black body, vestigial wings (alleles b, b, v g, v g).

F 1 dihybrid testcross: Wild type F 1 dihybrid female- gray body, normal wings (alleles b plus, b, v g plus, v g) x homozygous recessive male- black body, vestigial wings (alleles b, b, v g, v g). Testcross offspring: Cross of the F 1 s results in offspring that are wild type, black vestigial, gray vestigial and black normal. Testcross offspring: sperm are b v g. Eggs are b plus v g plus, b v g, b plus v g and b v g plus.

Results of fertilization: Sperm crossed with each egg in order above results in the following. Wild type, gray-normal (alleles b plus, b, v g plus, v g). Black vestigial (alleles b, b, v g, v g). Gray vestigial (alleles b plus, b, v g, v g). Black normal (alleles b, b, v g plus, v g).

Predicted ratios of testcross offspring. If genes are located on different chromosomes: 1 wild type gray normal to 1 black vestigial to 1 gray vestigial to 1 black normal. If genes are located on the same chromosome and parental alleles are always inherited together: 1 wild type gray normal to 1 black vestigial to 0 gray vestigial to 0 black normal. Results from Morgan's experiment: 965 wild type gray to 944 black vestigial to 206 gray vestigial to 0 black normal.

Conclusion

Since most offspring had a parental (P generation) phenotype, Morgan concluded that the genes for body color and wing size are genetically linked on the same chromosome. However, the production of a relatively small number of offspring with nonparental phenotypes indicated that some mechanism occasionally breaks the linkage between specific alleles of genes on the same chromosome.

Data from T. H. Morgan and C. J. Lynch, The linkage of two factors in *Drosophila* that are not sex-linked, *Biological Bulletin* 23:174–182 (1912).

WHAT IF? *If the parental (P generation) flies had been true-breeding for gray body with vestigial wings and black body with normal wings, which phenotypic class(es) would be largest among the testcross offspring?*

*For suggested answer, see*Appendix A*.*

The resulting flies had a much higher proportion of the combinations of traits seen in the P generation flies (called parental phenotypes) than would be expected if the two genes assorted independently. Morgan thus concluded that body color and wing size are usually inherited together in specific (parental) combinations because the genes are linked; they are near each other on the same chromosome:

> F 1 dihybrid female and homozygous recessive male in testcross. Body and wings alleles close together on same chromosomes. Most offspring are b plus b v g plus v g or b b v g v g.

F 1 dihybrid female and homozygous recessive male in testcross. Alleles for body and wings are shown close together on the same chromosomes.

Female: homologous chromosomes shown with alleles b plus (one homolog) and b (the other homolog) and v g plus (one homolog) and v g (the other homolog).

Male: homologous chromosomes with alleles b (one homolog) and b (the other homolog) and v g (one homolog) and v g (the other homolog). Most offspring are b plus b v g plus v g or b b v g v g.

As you proceed, be sure to keep in mind the distinction between the terms *linked genes* (two or more genes on the same chromosome that tend to be inherited together) and *sex-linked gene* (a single gene on a sex chromosome).

As Figure 15.9 shows, both of the combinations of traits not seen in the P generation (called nonparental phenotypes) were also produced in Morgan's experiments, suggesting that the body color and wing size alleles are not always linked genetically. To understand this conclusion, we need to further explore **genetic recombination**, the production of offspring with combinations of traits that differ from those found in either P generation parent.

Genetic Recombination and Linkage

Meiosis and random fertilization generate genetic variation among offspring of sexually reproducing organisms due to independent assortment of chromosomes, crossing over in meiosis I, and the possibility of any sperm fertilizing any egg (see Concept 13.4). Here we'll examine the chromosomal basis of recombination of alleles in relation to the genetic findings of Mendel and Morgan.

Recombination of Unlinked Genes: Independent Assortment of Chromosomes

Mendel learned from crosses in which he followed two characters that some offspring have combinations of traits that do not match those of either parent. For example, consider a cross of a dihybrid pea plant with yellow round seeds, heterozygous for both seed color and seed shape (YyRr), with a plant homozygous for both recessive alleles (with green wrinkled seeds, yyrr). (This cross acts as a testcross because the results will

reveal the genotype not only of the dihybrid YyRr plant, which we know, but of the gametes made in that plant.) Let's represent the cross by the following Punnett square:

Punnett square shows green wrinkled homozygous recessive parent (y y r r all lowercase) x yellow round dihybrid parent (uppercase Y lowercase y uppercase R lowercase r). Produces both parental-type and recombinant offspring.

Punnett square shows green wrinkled homozygous recessive parent (y y r r all lowercase) x yellow round dihybrid parent (uppercase Y lowercase y uppercase R lowercase r). Gametes from y y r r all lowercase parent are all yr both lowercase. Gametes from dihybrid parent are: YR both uppercase, yr both lowercase, uppercase Y lowercase r, and lowercase y uppercase R.

Results of fertilization of yr both lowercase gametes with dihybrid gametes listed above in order are as follows. Parental type offspring:

- Uppercase Y lowercase y uppercase R lowercase r (round yellow seed).
- y y r r r all lowercase (wrinkled green seed).

Recombinant offspring:

- Uppercase Y, y r r all lowercase (wrinkled yellow seed).
- y y both lowercase, uppercase R lowercase r (round green seed).

Notice in this Punnett square that one-half of the offspring are expected to inherit a phenotype that matches either of the phenotypes of the P (parental) generation originally crossed to produce the F1 dihybrid (see Figure 15.2). These matching offspring are called **parental types** (short for phenotypes). But two nonparental phenotypes are also found among the offspring. Because these offspring have new combinations of seed shape and color, they are called **recombinant types**, or **recombinants** for short. When 50% of all offspring are recombinants, as in this example, geneticists say that there is a 50% frequency of recombination. The predicted phenotypic ratios among the offspring are similar to what Mendel actually found in his YrRr×yyrr crosses.

A 50% frequency of recombination in such testcrosses is observed for any two genes that are located on different chromosomes and thus cannot be linked. The physical basis of recombination between unlinked genes is the random orientation of homologous chromosomes at metaphase I of meiosis, which leads to the independent assortment of the two unlinked genes (see Figure 13.11 and the question in the Figure 15.4 legend).

Recombination of Linked Genes: Crossing Over

Now, let's explain the results of the *Drosophila* testcross in Figure 15.9. Recall that most (83%) of the offspring from the testcross for body color and wing size had parental phenotypes. That suggested that the two genes were on the same chromosome, since the occurrence of parental types with a frequency greater than 50% indicates that the genes are linked. About 17% of offspring, however, were recombinants.

Seeing these results, Morgan proposed that some process must occasionally break the physical connection between specific alleles of genes on the same chromosome. Later experiments showed that this process, now called **crossing over**, accounts for the recombination of linked genes. In crossing over, which occurs while replicated homologous chromosomes are paired during prophase of meiosis I, a set of proteins breaks the DNA molecules of one maternal and one paternal chromatid and rejoins each to the other (see Figure 13.9). In effect, when a single crossover occurs, end portions of two nonsister chromatids trade places.

Figure 15.10 shows how crossing over in a dihybrid female fly resulted in recombinant eggs and ultimately recombinant offspring in Morgan's testcross. Most eggs had a chromosome with either the b+vg+ or bvg parental genotype, but some eggs had a recombinant chromosome (b+vg or bvg+). Fertilization of all classes of eggs by homozygous recessive sperm bvg produced an offspring population in which 17% exhibited a nonparental, recombinant phenotype, reflecting combinations of alleles not seen before in either P generation parent. In the **Scientific Skills Exercise**, you can use a statistical test to analyze the results from an F1 dihybrid testcross and see whether the two genes assort independently or are linked.

Figure 15.10

Chromosomal basis for recombination of linked genes.

Chromosomal basis for recombination of linked genes.

The diagram is as follows.

P generation (homozygous). Female wild type (gray body, normal wings) b plus b plus v g plus v g plus x male double mutant (black body, vestigial wings) b b v g v g. Female chromosomes are here shown as red, with the b plus / v g plus alleles close together on the same chromosomes. Male chromosomes are shown as pink, with the b/v g alleles close together on the same chromosomes.

F 1 dihybrid testcross. Wild type F 1 dihybrid (gray body, normal wings) b plus b v g plus v g with homozygous recessive male (black body, vestigial wings) b b v g v g. Female chromosomes are one red (b plus v g plus) and one pink (b v g). Male chromosomes are both blue (b v g and b v g). Each chromosome replicates so that there are two red maternal chromosomes joined as sister chromatids (both b plus v g plus) and two pink chromosomes joined as sister chromatids (both b v g). There are also two sets of sister chromatids for each blue chromosome (both b v g).

Meiosis I. Crossing over occurs so that maternal sister chromatids now show a pink section on one red chromatid and a red section on one of the pink chromatids. Red sister chromatids (one with pink section) now contain alleles b plus v g plus (same as before) on the all-red sister chromatid without crossing over, and on the red sister chromatid with pink section (crossing over) the alleles are now b plus v g. (The v g portion came from a pink chromosome).

Pink sister chromatids (one with red section from crossing over) now contain alleles b v g on the all pink chromatid (no crossing over) and b v g plus, with the v g plus in a red section from crossing over.

Meiosis I and II. Blue chromosomes from homozygous recessive male replicate to form 2 sets of sister chromatids, all with b v g alleles. Sperm all b v g on blue chromosome.

Punnett square shows testcross offspring. Following Meiosis II, half the eggs are parental types and recombinant. Sperm are b v g. Offspring are more often parental type than recombinant. Eggs made following Meiosis I from the female are:

- Parental type: red chromosome with b plus v g plus. Pink chromosome with b v g.
- Recombinant chromosomes: Red chromosome with pink section b plus v g. Pink chromosome with red section b v g plus.
- Crossing b v g sperm (blue chromosome) with each egg in the order listed above gives the following offspring. Parental types: 965 wild type (gray-normal) b plus v g plus (red chromosome and b v g (blue chromosome). 944 black-vestigial b v g (pink chromosome) b v g (blue chromosome). Recombinant offspring: 206 gray vestigial b plus v g (red chromosome with pink section), and b v g (blue chromosome). 185 black normal b v g plus (pink chromosome with red section) and b v g (blue chromosome).
- Recombination frequency = 391 recombinants divided by 2300 total offspring times 100 = 17%.

In these diagrams re-creating the testcross in Figure 15.9, we track chromosomes as well as genes. The maternal chromosomes (those present in the wild-type F1 dihybrid) are color-coded red and pink to distinguish one homolog from the other before any meiotic crossing over has occurred. Because crossing over between the b+/b and vg+/vg loci occurs in some, but not most, egg-producing cells, more eggs with parental-type chromosomes than with recombinant ones are produced in the mating females. Fertilization of the eggs by sperm of genotype bvg gives rise to some recombinant offspring. The recombination frequency is the percentage of recombinant flies in the total pool of offspring.

DRAW IT *Suppose, as in the question at the bottom of*Figure 15.9*, the parental (*P*generation) flies were true-breeding for gray body with vestigial wings and black body with normal wings. Draw the chromosomes in each of the four possible kinds of eggs from an*F1*female, and label each chromosome as “parental” or “recombinant.”*

*For suggested answer, see*Appendix A.**

Animation

Linked Genes and Crossing Over

image

Play

image

Play

Scientific Skills Exercise

Using the Chi-Square(χ2)Test

AP Science Practices 5, 6

Are Two Genes Linked or Unlinked? Genes that are in close proximity on the same chromosome will result in the linked alleles being inherited together more often than not. But how can you tell if certain alleles are inherited together due to linkage or whether they just happen to assort together randomly? In this exercise, you'll use a simple statistical test, the chi-square (χ2) test, to analyze phenotypes of F1 testcross progeny in order to see whether two genes are linked or unlinked.

How These Experiments Are Done If genes are unlinked and assorting independently, the phenotypic ratio of offspring from an F1 testcross is expected to be 1:1:1:1 (see Figure 15.9). If the two genes are linked, however, the observed phenotypic ratio of the offspring will not match that ratio. Given that random fluctuations in the data do occur, how much must the observed numbers deviate from the expected numbers for us to conclude that the genes are not assorting independently but may instead be linked?

To answer this question, scientists use a statistical test. This test, called a chi-square (χ2) test, compares an observed data set with an expected data set predicted by a hypothesis (here, that the genes are unlinked) and measures the discrepancy between the two, thus determining the "goodness of fit." If the discrepancy between the observed and expected data sets is so large that it is unlikely to have occurred by random fluctuation, we say there is statistically significant evidence against the hypothesis (or, more specifically, evidence for the genes being linked). If the discrepancy is small, then our observations are well explained by random variation alone. In this case, we say that the observed data are consistent with our hypothesis, or that the discrepancy is statistically insignificant. Note, however, that consistency with our hypothesis is not

the same as proof of our hypothesis. Also, the size of the experimental data set is important: With small data sets like this one, even if the genes are linked, discrepancies might be small by chance alone if the linkage is weak. For simplicity, we overlook the effect of sample size here.

Cosmos plants.

Cosmos plants

Data from the Simulated Experiment In cosmos plants, purple stem (A) is dominant to green stem (a), and short petals (B) is dominant to long petals (b). In a simulated cross, AABB plants were crossed with aabb plants to generate F1 dihybrids (AaBb), which were then testcrossed (AaBb×aabb). A total of 900 offspring plants were scored for stem color and flower petal length.

A table is given.

- If the phenotype is dominant, a dash is used for the second allele; it could be either the dominant or recessive allele.

The table is as follows:

Offspring from testcross of upper A, lower a, upper B, lower b (F 1) crossed with lower a, a, b, b	Purple stem slash short petals (upper A dash upper B dash)	Green stems slash short petals (lower a, a, upper B dash)	Purple stem slash long petals (upper A, dash, lower b b)	Green stem slash long petals (lower a, a, b, b)
Expected ratio if the genes are unlinked	1	1	1	1
Expected number of offspring (of 900)				
Observed number of offspring (of 900)	220	210	231	239

INTERPRET THE DATA

- The results in the data table are from a simulated F1 dihybrid testcross. The hypothesis that the two genes are unlinked predicts that the offspring phenotypic ratio will be 1:1:1:1. Using this ratio, calculate the expected number of each phenotype out of the 900 total offspring, and enter the values in that data table.

- The goodness of fit is measured by χ2. This statistic measures the amounts by which the observed values differ from their respective predictions to indicate how closely the two sets of values match. The formula for calculating this value is

χ2=Σ(o−e)2e

- where Σ=sum of, o=observed and e=expected*.* Calculate the χ2 value for the data using the table below. Fill out that table, carrying out the operations indicated in the top row. Then add up the entries in the last column to find the χ2 value.

A table is given.

The table is as follows:

Testcross offspring	Expected (e)	Observed (o)	Deviation (o minus e)	left parenthesis, o minus e, right parenthesis, squared	left parenthesis, o minus e, right parenthesis, squared, divided by e
upper A, dash, upper B, dash		220			
lower a, a, upper B, dash		210			
upper A, dash, lower b, b		231			
lower a, a, b, b		239			
Chi squared equals sum					

- The χ2 value means nothing on its own—it is used to find the probability that, assuming the hypothesis is true, the observed data set could have resulted from random fluctuations. A low probability suggests that the observed data are not consistent with the hypothesis and thus the hypothesis should be rejected. A standard cutoff point used by biologists is a probability of 0.05

(5%). If the probability corresponding to the χ^2 value is 0.05 or less, the differences between observed and expected values are considered statistically significant and the hypothesis (that the genes are unlinked) should be rejected. If the probability is above 0.05, the results are not statistically significant; the observed data are consistent with the hypothesis that the genes are unlinked.

- To find the probability, locate your χ^2 value in the χ^2 distribution table in Appendix D. The "degrees of freedom" (df) of your data set is the number of categories (here, 4 phenotypes) minus 1, so df=3.
-
    - Determine which values on the df=3 line of the table your calculated χ^2 value lies between.
- The column headings for these values show the probability range for your χ^2 number. Based on whether there are nonsignificant ($p>0.05$) or significant ($p\leq0.05$) differences between the observed and expected values, are the data consistent with the hypothesis that the two genes are unlinked and assorting independently, or is there enough evidence to reject this hypothesis?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

New Combinations of Alleles: Variation for Natural Selection

EVOLUTION The physical behavior of chromosomes during meiosis contributes to the generation of variation in offspring (see Concept 13.4). Each pair of homologous chromosomes lines up independently of other pairs during metaphase I, and crossing over prior to that, during prophase I, can mix and match parts of maternal and paternal homologs. Mendel's elegant experiments show that the behavior of the abstract entities known as genes—or, more concretely, alleles of genes—also leads to variation in offspring (see Concept 14.1). Now, putting these different ideas together, you can see that the recombinant chromosomes resulting from crossing over may bring alleles together in new combinations, and the subsequent events of meiosis distribute to gametes the recombinant chromosomes in a multitude of combinations, such as the new variants seen in Figures 15.9 and 15.10. Random fertilization then increases even further the number of variant allele combinations that can be created.

This abundance of genetic variation provides the raw material on which natural selection works. If the traits conferred by particular combinations of alleles are better suited for a given environment, organisms possessing those genotypes will be expected to thrive and leave more offspring, ensuring the continuation of their genetic complement. In the next generation, of course, the alleles will be shuffled anew. Ultimately, the interplay between environment and phenotype (and thus genotype) will determine which genetic combinations persist over time.

Mapping the Distance Between Genes Using Recombination Data: *Scientific Inquiry*

The discovery of linked genes and recombination due to crossing over motivated one of Morgan's students, Alfred H. Sturtevant, to work out a method for constructing a **genetic map**, an ordered list of the genetic loci along a particular chromosome.

Sturtevant hypothesized that the percentage of recombinant offspring, the *recombination frequency*, calculated from experiments like the one in Figures 15.9 and 15.10, depends on the distance between genes on a chromosome. He assumed that crossing over is a random event, with the chance of crossing over approximately equal at all points along a chromosome. Based on these assumptions, Sturtevant predicted that *the farther apart two genes are, the higher the probability that a crossover will occur between them and therefore the higher the recombination frequency*. His reasoning was simple: The greater the distance between two genes, the more points there are between them where crossing over can occur. Using recombination data from various fruit fly crosses, Sturtevant proceeded to assign relative positions to genes on the same chromosomes—that is, to *map* genes.

A genetic map based on recombination frequencies is called a **linkage map**. **Figure 15.11** shows Sturtevant's linkage map of three genes: the body color (b) and wing size (vg) genes depicted in Figure 15.10 and a third gene, called cinnabar (cn). Cinnabar is one of many *Drosophila* genes affecting eye color. Cinnabar eyes, a mutant phenotype, are a brighter red than the wild-type color. The recombination frequency between cn and b is 9%; that between cn and vg, 9.25%; and that between b and vg, 17%. In other words, crossovers between cn and b and between cn and vg are about half as frequent as crossovers between b and vg. Only a map that locates cn about midway between b and vg is consistent with these data, as you can prove to yourself by drawing alternative maps. Sturtevant expressed the distances between genes in **map units**, defining one map unit as equivalent to a 1% recombination frequency.

Figure 15.11

Research Method: Constructing a Linkage Map

Application

A linkage map shows the relative locations of genes along a chromosome.

Technique

A linkage map is based on the assumption that the probability of a crossover between two genetic loci is proportional to the distance separating the loci. The recombination frequencies used to construct a linkage map for a particular chromosome are obtained from experimental crosses, such as the cross depicted in Figures 15.9 and 15.10. The distances between genes are expressed as map units, with one map unit equivalent to a 1% recombination frequency. Genes are arranged on the chromosome in the order that best fits the data.

Results

In this example, the observed recombination frequencies between three *Drosophila* gene pairs (b and cn, 9%; cn and vg, 9.5%; and b and vg, 17%) best fit a linear order in which cn is positioned about halfway between the other two genes:

A diagram of recombination frequencies shows c n positioned approximately halfway between b to the left and v g to the right. from b to v g = 17%. From b to c n = 9%. From c n to v g = 9.5%.

The b-vg recombination frequency (17%) is slightly less than the sum of the b-cn and cn-vg frequencies (9+9.5=18.5%) because of the few times that one crossover occurs between b and cn and another crossover occurs between cn and vg. The second crossover would “cancel out” the first, reducing the observed b-vg recombination frequency while contributing to the frequency between each of the closer pairs of genes. The value of 18.5% (18.5 map units) is closer to the actual distance between the genes. In practice, a geneticist would add the smaller distances in constructing a map.

In practice, the interpretation of recombination data is more complicated than this example suggests. Some genes on a chromosome are so far from each other that a crossover between them is virtually certain. The observed frequency of recombination in crosses involving two such genes can have a maximum value of 50%, a result indistinguishable from that for genes on different chromosomes. In this case, the physical connection between genes on the same chromosome is not reflected in the results of genetic crosses. Despite being on the same chromosome and thus being *physically connected*, the genes are *genetically unlinked*; alleles of such genes assort independently, as if they were on different chromosomes. In fact, at least two of the genes for pea characters that Mendel studied are now known to be on the same chromosome, but the distance between them is so great that linkage is not observed in genetic crosses. Consequently, the two genes behaved as if they were on different chromosomes in Mendel’s experiments. Genes located far apart on a chromosome are mapped by adding the recombination frequencies from crosses involving closer pairs of genes lying between the two distant genes.

Using recombination data, Sturtevant and his colleagues were able to map numerous *Drosophila* genes in linear arrays. They found that the genes clustered into four groups of linked genes (*linkage groups*). Light microscopy had revealed four pairs of chromosomes in *Drosophila*, so the linkage map provided additional

evidence that genes are located on chromosomes. Each chromosome has a linear array of specific genes, each gene with its own locus **(Figure 15.12)**.

Figure 15.12

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    A partial genetic (linkage) map of a *Drosophila*
chromosome.
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Linkage map of 7 genes on a homolog of Drosophila chromosome 2; pull out diagram also shows chromosome pairs 1, 3, 4 and X Y.

Genes drawn at map unit distances from arista length from left to right:

0: arista length (appendages on head). Mutant phenotype is short aristae and wild type is long. 16.5: eye color. Mutant phenotype is maroon eyes and wild type is red eyes.

48.5: body color. Mutant phenotype is black and wild type is gray.

57.5: eye color. Mutant phenotype is cinnabar and wild type is red.

67.0: wings. Mutant phenotype is vestigial and wild type is normal.

75.5: wing shape. Mutant phenotype is down-curved wings and wild type is normal.

104.5: eye color. Mutant phenotype is brown and wild type is red.

This partial map shows just seven mapped genes on *Drosophila* chromosome II. (DNA sequencing has revealed over 9,000 genes on that chromosome.) The number at each gene locus is the number of map units from the arista length gene (far left).

Because a linkage map is based strictly on recombination frequencies, it gives only an approximate picture of a chromosome. The frequency of crossing over is not actually uniform over the length of a chromosome, as Sturtevant assumed, and therefore map units do not correspond to actual physical distances (in nanometers, for instance). A linkage map does portray the order of genes along a chromosome, but it does not accurately portray the precise locations of those genes. Other methods enable geneticists to construct *cytogenetic maps* of chromosomes, which locate genes with respect to chromosomal features, such as stained bands, that can be seen in the microscope. Technical advances over recent decades have enormously increased the rate and affordability of DNA sequencing. Therefore, today, most researchers sequence whole genomes to map the locations of genes of a given species. The entire nucleotide sequence is the ultimate physical map of a chromosome, revealing the physical distances in DNA nucleotides between gene loci (see Concept 21.1).

Comparing a linkage map with such a physical map or with a cytogenetic map of the same chromosome, we find that the linear order of genes is identical in all the maps, but the spacing between genes is not.